

Mach-Zehnder Interferometer Design Proposal

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Abstract—This proposal describes the design of a low loss Mach-Zehnder interferometer on a silicon on insulator (SOI) platform. The design considers waveguide loss and taper lengths used for varying waveguide transitions. Analytical and simulated results will be used to evaluate different design parameters and guide the selection of the interferometer design.

I. INTRODUCTION

Mach-Zehnder interferometers (MZIs) are commonly used in integrated photonics for optical signal processing, optical control systems, and other optical functions [1]. Implementing an MZI on a silicon on insulator (SOI) platform enables multiple photonic components to be integrated onto a single chip while maintaining a compact device footprint [2], [3]. The high refractive index contrast between silicon and silicon dioxide provides strong optical confinement within the waveguide, enabling compact waveguide structures and dense integration of photonic components [1]. In this proposal, the effects of waveguide dimensions and taper length on the interference behavior and loss of the MZI will be investigated. Analytical and simulated results will be used to evaluate these effects and assist with the final design of the device.

II. THEORY

A Mach-Zehnder interferometer operates by splitting an input optical signal into two paths and recombining the signals. The relative phase between the two optical paths determines the output intensity of the MZI. An integrated photonic MZI consists of an input splitter, two waveguide arms, and an output combiner. A difference in the lengths of the two arms produces a difference in optical path length, defined as [4]

$$\Delta L = |L_1 - L_2|. \quad (1)$$

For a balanced MZI, the two arms have equal lengths and $\Delta L = 0$. For an unbalanced MZI, $\Delta L \neq 0$, resulting in a difference in the phase due to the optical fields propagating through the two arms of the MZI. When the two fields are recombined, they interfere constructively or destructively depending on their relative phase. Since the phase difference depends on wavelength, an unbalanced MZI produces a wavelength dependent interference response.

To describe the optical fields within the MZI, consider an ideal 50:50 splitter with input intensity I_i and electric field E_i . The splitter divides the input power equally between the two arms, giving

$$I_1 = I_2 = \frac{I_i}{2}. \quad (2)$$

Since the optical intensity is proportional to the magnitude squared of the electric field, the electric field in each arm is

$$E_1 = E_2 = \frac{E_i}{\sqrt{2}}. \quad (3)$$

As the fields propagate through the two arms, they accumulate both phase and propagation loss. The fields at the output of the two arms can be written as

$$E_{o1} = \frac{E_i}{\sqrt{2}} e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} \quad (4)$$

and

$$E_{o2} = \frac{E_i}{\sqrt{2}} e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2}, \quad (5)$$

where β_1 and β_2 are the propagation constants, α_1 and α_2 are the power attenuation coefficients, and L_1 and L_2 are the lengths of the two MZI arms [4].

For an ideal 50:50 combiner, the electric field at the selected output port is

$$E_o = \frac{1}{\sqrt{2}} (E_{o1} + E_{o2}), \quad (6)$$

which gives

$$E_o = \frac{E_i}{2} \left[e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \right]. \quad (7)$$

The output intensity is proportional to the magnitude squared of the total electric field,

$$I_o = |E_o|^2. \quad (8)$$

For the ideal case in which the two arms have equal propagation loss, the transfer function can be written as [4]

$$\frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(\Delta\phi)], \quad (9)$$

where the phase difference between the two arms is

$$\Delta\phi = \beta_1 L_1 - \beta_2 L_2. \quad (10)$$

The output intensity depends on the relative phase between the two optical paths. When $\Delta\phi$ is an integer multiple of 2π , the fields interfere constructively and the output intensity is maximized. When $\Delta\phi$ is an odd multiple of π , the fields interfere destructively and the output intensity is minimized [4].

The propagation constant is related to the effective index of the waveguide by

$$\beta = \frac{2\pi n_{\text{eff}}}{\lambda}. \quad (11)$$

For an MZI with identical waveguide geometries in both arms, the phase difference can therefore be related to the path length difference by

$$\Delta\phi = \frac{2\pi n_{\text{eff}} \Delta L}{\lambda}. \quad (12)$$

The periodicity of the MZI transmission spectrum is described by the free spectral range (FSR), which is the wavelength spacing between transmission maxima or minima. For an MZI with a path length difference ΔL , the wavelength domain FSR is [4]

$$\text{FSR}(\lambda) = \frac{\lambda^2}{n_g \Delta L}, \quad (13)$$

where n_g is the group index of the waveguide. The group index is related to the effective index by [4]

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (14)$$

The waveguide dimensions can influence the MZI spectral response through their effect on the effective index and the group index. Changes in the width of the waveguide can alter n_{eff} and n_g , which in turn affect the phase difference and the FSR. These effects are particularly important for this design because the proposed MZIs use different waveguide geometries and taper transitions.

III. MODELING AND SIMULATION

A. Waveguide Modeling

The first step in the design was to characterize the optical properties of the waveguides that will be used in the MZI. Two waveguide geometries were investigated: a 500 nm wide by 220 nm high strip waveguide and a 3 μm wide by 220 nm high waveguide. 220 nm waveguide height is commonly used at foundries [4]. The waveguide structures were modeled using Lumerical MODE to determine the supported optical modes and wavelength dependent effective indices [5].

For the 500 nm wide waveguide, the fundamental quasi-transverse electric (TE) and quasi-transverse magnetic (TM) modes were investigated. The simulated electric field intensity distributions for the TE and TM modes are shown in Figures 1 and 2, respectively. The 3 μm wide waveguide supports multiple modes due to its larger cross sectional dimensions. Therefore, the fundamental mode was selected for the analysis of the 3 μm wide waveguide, as shown in Figure 3.

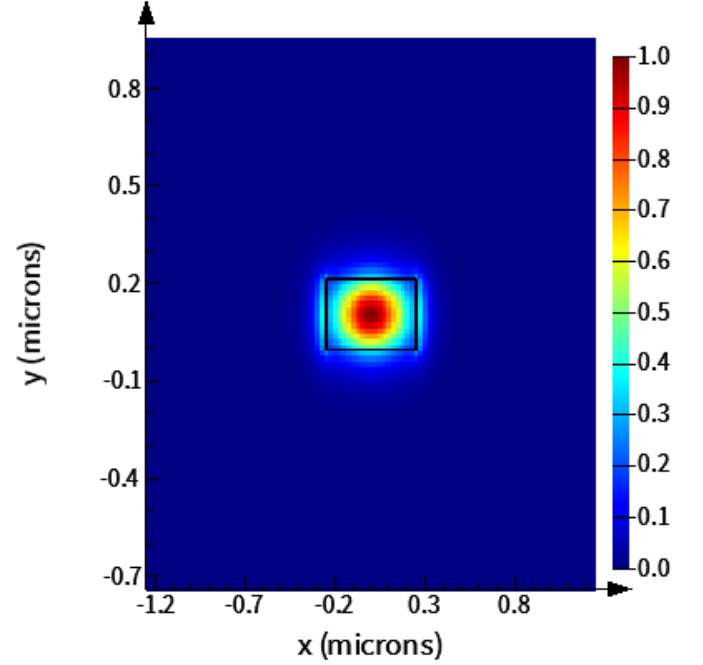


Fig. 1. Simulated electric field intensity of the fundamental quasi-TE mode for the 500 nm $\times$ 220 nm waveguide.

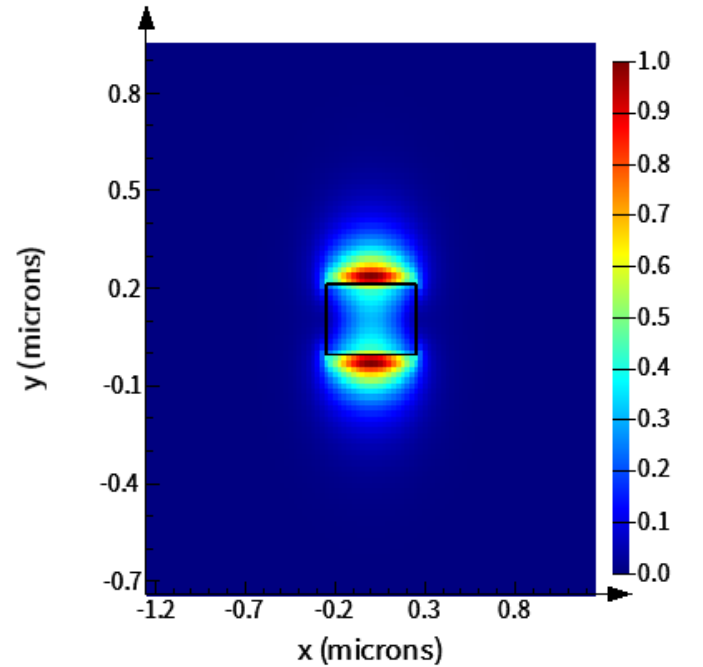


Fig. 2. Simulated electric field intensity of the fundamental quasi-TM mode for the 500 nm $\times$ 220 nm waveguide.

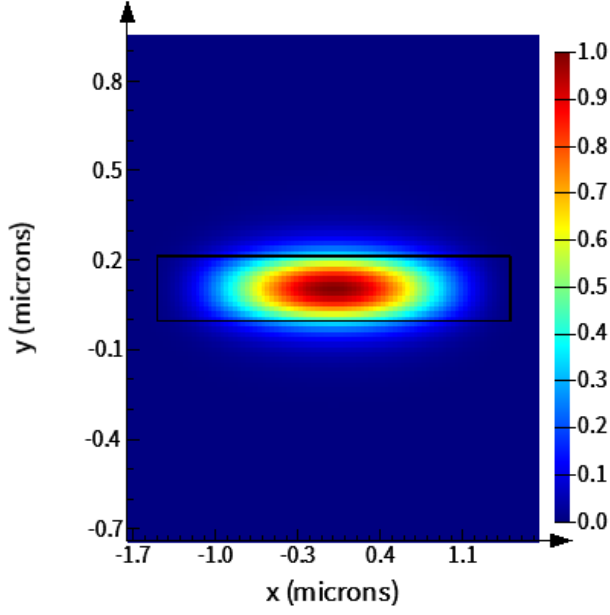


Fig. 3. Simulated electric field intensity of the fundamental mode for the $3\ \mu\text{m} \times 220\ \text{nm}$ waveguide.

A wavelength sweep from 1500 nm to 1600 nm was performed to determine the effective index of each waveguide. The 500 nm wide waveguide was analyzed using the fundamental quasi-TE mode, while the fundamental mode was used for the $3\ \mu\text{m}$ wide waveguide. The effective index and group index of the 500 nm $\times$ 220 nm waveguide are shown in Figures 4 and 5, respectively. The corresponding results for the $3\ \mu\text{m} \times 220\ \text{nm}$ waveguide are shown in Figures 6 and 7.

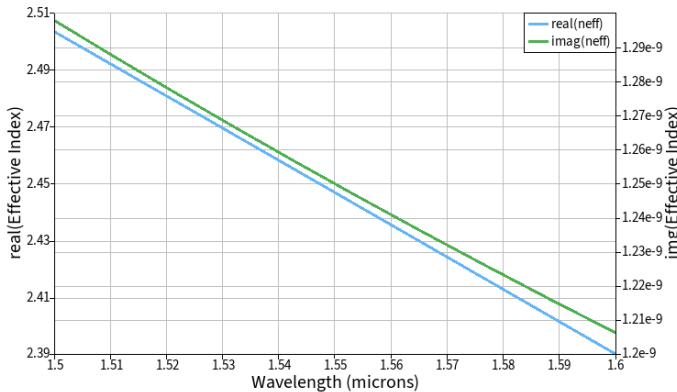


Fig. 4. Effective index as a function of wavelength for the 500 nm $\times$ 220 nm waveguide.

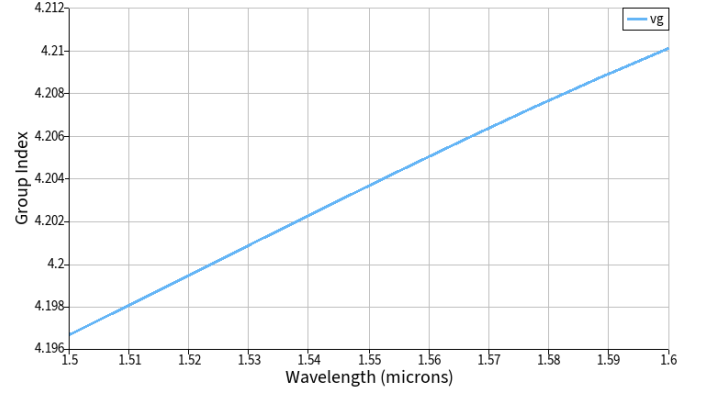


Fig. 5. Group index as a function of wavelength for the 500 nm $\times$ 220 nm waveguide.

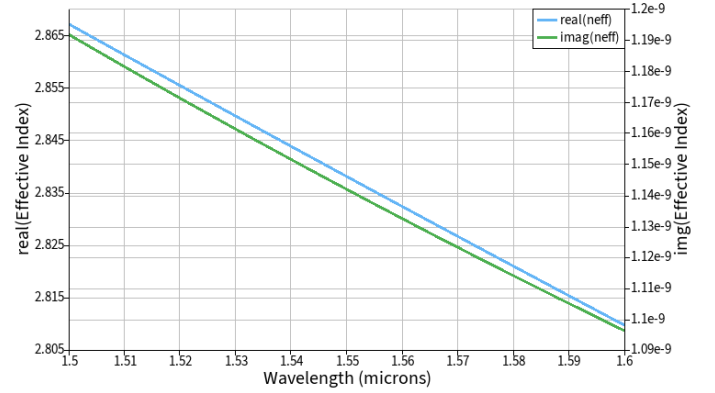


Fig. 6. Effective index as a function of wavelength for the $3\ \mu\text{m} \times 220\ \text{nm}$ waveguide.

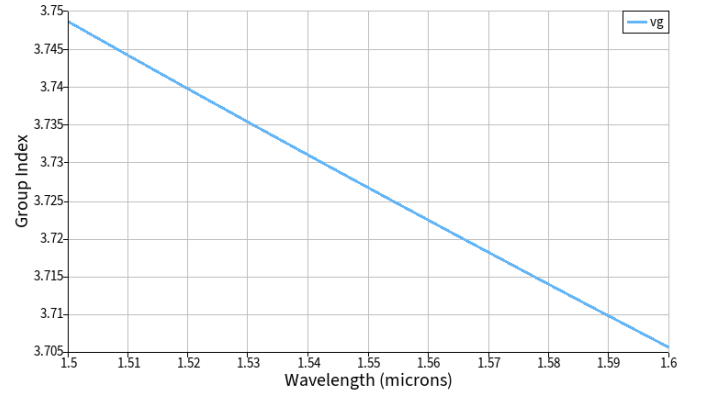


Fig. 7. Group index as a function of wavelength for the $3\ \mu\text{m} \times 220\ \text{nm}$ waveguide.

The simulation results show that the effective index decreases with increasing wavelength for both waveguide geometries, and the group index increases over the wavelength range. These wavelength dependent properties are important for predicting the phase and free spectral range of the MZI.

To simplify the waveguide models for use in the circuit simulation, the simulated effective index data were fitted using

a second order Taylor expansion about the center wavelength, $\lambda_0 = 1.55 \mu\text{m}$. The effective index is [4]

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2, \quad (15)$$

where n_1 , n_2 , and n_3 are fitting coefficients and wavelength is expressed in micrometers.

The resulting fit for the $500 \text{ nm} \times 220 \text{ nm}$ waveguide is shown in Figure 8. The resulting compact model is

$$n_{\text{eff}}(\lambda) = 2.44682 - 1.13339(\lambda - 1.55) - 0.0439366(\lambda - 1.55)^2. \quad (16)$$

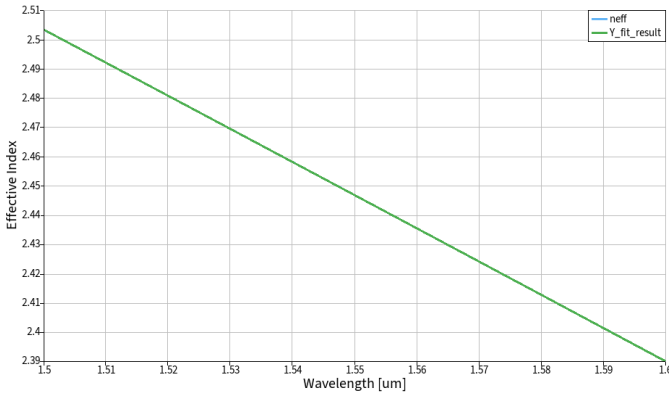


Fig. 8. Second order polynomial fit of the simulated effective index for the $500 \text{ nm} \times 220 \text{ nm}$ waveguide.

For the $3 \mu\text{m} \times 220 \text{ nm}$ waveguide, the corresponding fit is shown in Figure 9, giving

$$n_{\text{eff}}(\lambda) = 2.83808 - 0.573412(\lambda - 1.55) + 0.138567(\lambda - 1.55)^2. \quad (17)$$

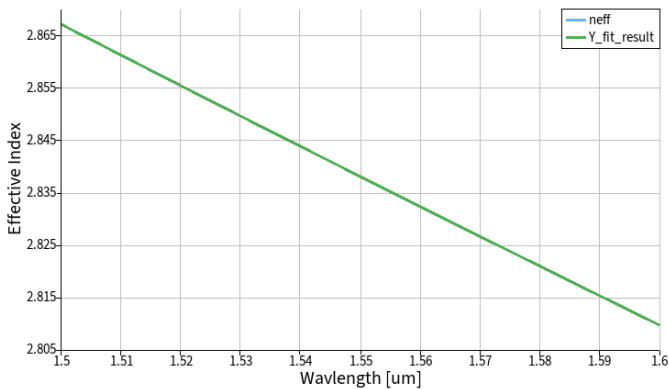


Fig. 9. Second order polynomial fit of the simulated effective index for the $3 \mu\text{m} \times 220 \text{ nm}$ waveguide.

These polynomial expressions provide compact models of the wavelength dependent effective index and can be used to calculate the propagation constant and group index without repeatedly performing the full mode simulation.

B. MZI Modeling

After characterizing the individual waveguides, the complete MZI was modeled using Lumerical INTERCONNECT. The circuit level model was developed to evaluate the transmission response of the MZI and determine how the waveguide geometry affects the interference pattern [4], [6].

The simulated optical circuit includes grating couplers, Y-branch splitters and combiners, and the two MZI waveguide arms. The grating couplers provide the interface between the optical input/output and the integrated waveguide. A simulated grating coupler response is shown in Figure 10. The input light is divided between the two MZI arms using a Y-branch splitter. The loss associated with an individual Y-branch is shown in Figure 11.

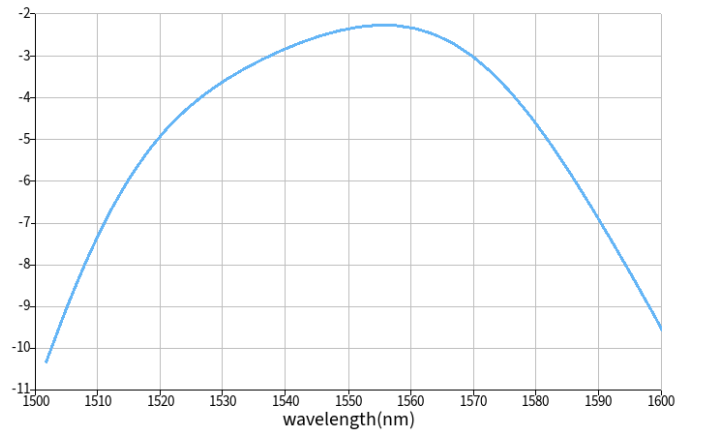


Fig. 10. Simulated coupling response of a single grating coupler.

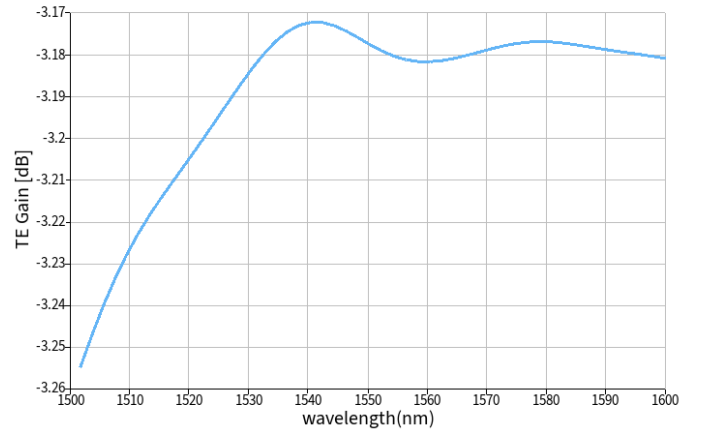


Fig. 11. Simulated transmission and loss of a single Y-branch.

The two MZI arms were designed with lengths of 2.1 mm and 0.1 mm, resulting in a geometric path length difference of

$$\Delta L = |2.1 - 0.1| \text{ mm} = 2.0 \text{ mm}. \quad (18)$$

The two optical paths are recombined with a second Y-branch. Two waveguide configurations were investigated. The

first configuration uses a 500 nm wide waveguide throughout the MZI. The second configuration uses a 3 μm wide waveguide for the straight sections, with the waveguide transitioning to a 500 nm width in the bent sections. The transitions between the two widths will be with implemented 25 μm tapers.

The taper transitions were evaluated separately to determine whether their loss would significantly affect the circuit level MZI response. As shown in Figure 12, the simulated taper loss is small over the range of taper lengths considered. The taper transitions were not included explicitly in the INTERCONNECT circuit model. The taper geometry will instead be included during the layout and fabrication stage using KLayout.

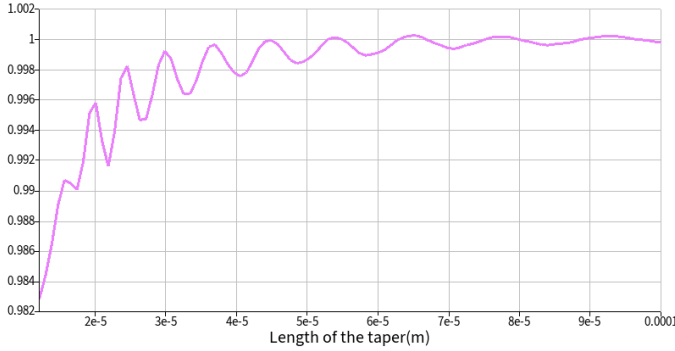


Fig. 12. Simulated taper loss as a function of taper length.

The difference in group index between the two waveguide geometries also affects the expected free spectral range. The wavelength domain FSR for identical waveguide geometries can be calculated using (13).

Because the two waveguide widths have different group indices, the expected FSR changes even when the path length difference remains constant. The calculated FSR values for the two waveguide configurations are summarized in Table I.

TABLE I
GROUP INDEX AND CALCULATED FSR DEPENDENT ON WAVELENGTH FOR THE TWO WAVEGUIDE GEOMETRIES AT $\lambda = 1550$ nm AND $\Delta L = 2.0$ mm.

Waveguide Width (μm)	n_g	FSR (nm)
3.0	3.7267	0.3223
0.5	4.20367	0.2858

The complete MZI circuit models are shown in Figure 13. The simulated transmission spectra for the two waveguide configurations are shown in Figure 14. The transmission spectra demonstrate the wavelength dependent interference produced by the difference in optical path length between the two MZI arms. Differences in the spectral response between the two configurations result from the different group indices associated with the waveguide geometries.

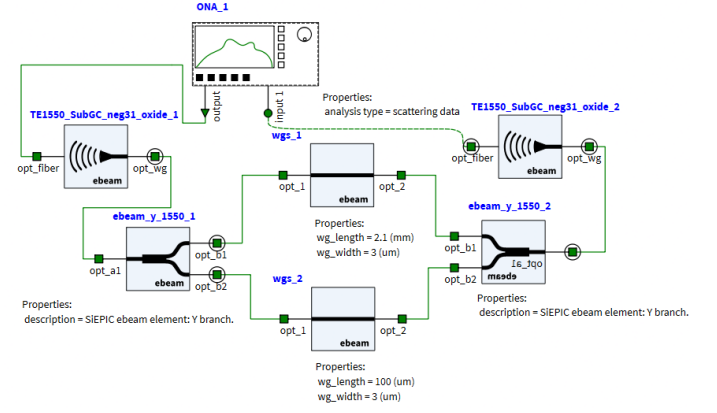


Fig. 13. Lumerical INTERCONNECT layout of the complete MZI circuit.

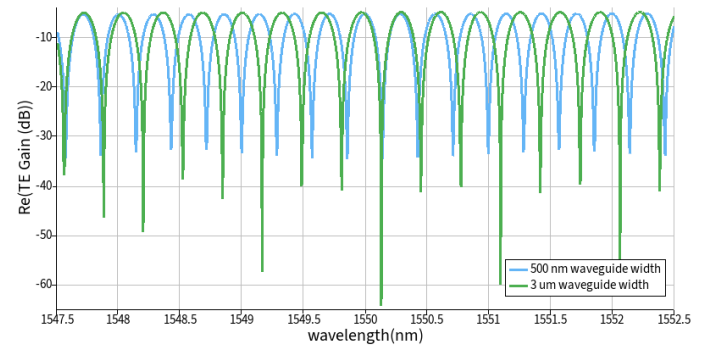


Fig. 14. Simulated transmission spectra of the MZIs using the 500 nm and 3 μm waveguide geometries.

The modeling process establishes a connection between the waveguide dimensions and the expected MZI response. The MODE simulations provide the effective index and group index data required to describe the individual waveguides. This provides a basis for selecting the waveguide geometry and predicting the expected spectral response before fabrication.

IV. CONCLUSION

This report demonstrates the design and simulation of a silicon on insulator Mach-Zehnder interferometer with an emphasis on the effects of waveguide geometry on the optical response. Two waveguide configurations were investigated: a 500 nm $\times$ 220 nm strip waveguide and a 3 μm $\times$ 220 nm waveguide.

Lumerical MODE simulations were used to determine the supported modes and wavelength dependent effective and group indices of the two waveguide geometries [5]. The simulated effective index data is represented using second order polynomial models about a center wavelength of 1550 nm. These compact models provide a practical method for incorporating the wavelength dependent properties of the waveguides into circuit level simulations.

The waveguide models were subsequently implemented in Lumerical INTERCONNECT to simulate the complete MZI [6]. The circuit included the grating couplers, Y-branch splitter, the two unequal length waveguide arms, and a Y-branch

combiner. The resulting transmission spectra demonstrate the wavelength dependent interference expected from the 2.0 mm path length difference. The difference in group index between the two waveguide geometries also results in a difference in the expected free spectral range.

The taper analysis indicates that the transition between the 3 μm and 500 nm waveguides introduces small loss for the taper to be ignored in the circuit level INTERCONNECT model. The taper geometry will be included in the final layout and evaluated during fabrication and experimental characterization.

The modeling results provide a basis for the fabrication and experimental investigation of the proposed MZIs. Comparing the fabricated devices will allow the effects of waveguide width, propagation loss, bend loss, and taper transitions to be evaluated experimentally. The combined simulation and experimental results can then be used to determine how the waveguide geometry influences the optical performance of an integrated silicon MZI.

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